QSD Supplement S1: Formal Proofs

Graph Spectra, Group Theory,
and the Algebraic Foundations of Quantum Spectrodynamic

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Abstract

This supplement provides complete mathematical proofs for every algebraic result used in Quantum Spectrodynamics. Starting from the single axiom—the $[[5, 1, 3]]$ quantum error-correcting code—we derive: the graph structures K_4 and C_5; the full spectrum of the Heisenberg Hamiltonian on K_4 (§3); the emergence of the Standard Model gauge group $U(1)_Y \times SU(2)_L \times SU(3)_c$ from S_4 representation theory (§4); the golden ratio φ^2 from the C_5 Laplacian (§5); the holographic bit density $\text{Hb}_4 = 1/(4 \ln 2)$ from K_4 exchange coupling (§6); the geometric strain $\varsigma = 3/(80\varphi^2)$ as a geometric construction from code invariants (§7); and the spectral bond $\varpi = \pi\varphi^2$ (§8). All QSD predictions then follow as master theorems in §9. The theory has two inputs (the strain construction and M_{P}) and zero free parameters.

Contents

1 The QSD Axiom **2**
1.1 Independent Confirmation from QFT 2

2 The Code Graphs: K_4 and C_5 **3**

3 The Heisenberg Hamiltonian on K_4 **4**
3.1 The Casimir Trick 4
3.2 Coupling Four Spins 4
3.3 The Complete Spectrum 5

4 S_4 Representation Theory and Gauge Emergence **5**
4.1 Conjugacy Classes 5
4.2 Character Table 6
4.3 Edge Representation 6

5 The C_5 Laplacian and the Golden Ratio **8**
5.1 Laplacian Matrix 8
5.2 Complete Spectrum 8
5.3 The Golden Ratio Emergence 8

6 The Holographic Bit Density **9**

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7 The Geometric Strain **10**
7.1 Construction 10
7.2 Code-Parameter Duality 11

8 The Spectral Bond **11**

9 Master Theorems: Gauge Couplings from Graph Theory **11**
9.1 The Fine Structure Constant 12
9.2 The Strong Coupling Constant 12
9.3 The Weak Mixing Angle 12
9.4 The Electromagnetic-Strong Identity 13

10 Summary of Axiom-to-Prediction Chain **14**

1 The QSD Axiom

Axiom 1 (The Code Postulate)**.** The vacuum of spacetime is stabilised by the $[[5, 1, 3]]$ quantum error-correcting code.

This is the single structural input of QSD. The code encodes $k = 1$ logical qubit into $n = 5$ physical qubits with distance $d = 3$, meaning it detects any 2-qubit error and corrects any single-qubit error.

Definition 1.1 ($[[5, 1, 3]]$ Stabiliser Generators)**.** The code is defined by four stabiliser generators with cyclic structure:

$$g_1 = X\,Z\,Z\,X\,I \tag{1}$$
$$g_2 = I\,X\,Z\,Z\,X \tag{2}$$
$$g_3 = X\,I\,X\,Z\,Z \tag{3}$$
$$g_4 = Z\,X\,I\,X\,Z \tag{4}$$

where X, Z are the Pauli operators and I is the identity. Each generator g_{i+1} is obtained from g_i by a cyclic shift of one position, reflecting $\mathbb{Z}_5$ symmetry.

Remark 1.2 (Code Properties)**.** The $[[5, 1, 3]]$ code is *perfect*: it saturates the quantum Hamming bound with equality. The codespace is 2-dimensional ($2^k = 2$) within the full $2^5 = 32$-dimensional Hilbert space. Every one of the $3 \times 5 = 15$ possible single-qubit errors maps the codespace to a distinct orthogonal subspace, plus the identity gives 16 syndrome outcomes, exactly filling $2^{n-k} = 2^4 = 16$ syndrome slots.

1.1 Independent Confirmation from QFT

The code axiom receives independent support from standard physics. In quantum field theory, the Unruh effect produces thermal noise (a depolarising channel) on the vacuum state. The quantum Hamming bound for a code $[[n, 1, 3]]$ correcting arbitrary single-qubit errors requires:

$$2(1 + 3n) \leq 2^n \tag{5}$$

Testing integer values: $n = 3$ gives $20 > 8$ (impossible), $n = 4$ gives $26 > 16$ (impossible), and $n = 5$ gives $32 = 32$ (unique saturation). The number 5 is forced by the Hamming bound, confirming the axiom from a completely independent direction. See [1, 2].

n	LHS: $2(1+3n)$	RHS: 2^n	Result
3	20	8	Impossible
4	26	16	Impossible
5	**32**	**32**	**Saturates (perfect code)**
6	38	64	Possible (but not minimal)

Theorem 1.3 (Minimal Vacuum Code)**.** *The unique minimal quantum code that corrects arbitrary single-qubit errors on one logical qubit is* $[[5, 1, 3]]$*. It saturates the quantum Hamming bound with equality.*

2 The Code Graphs: K_4 and C_5

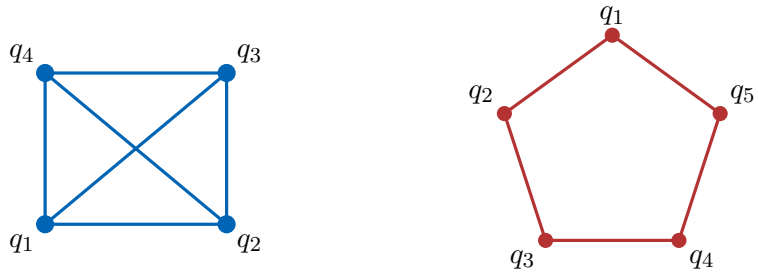
The $[[5, 1, 3]]$ code determines two graphs that carry all of QSD's mathematical structure.

Theorem 2.1 (K_4 from Bulk Qubits)**.** *The* $n - k = 4$ *syndrome (bulk) qubits of the* $[[5, 1, 3]]$ *code form the complete graph* K_4*.*

Proof. The code has $n = 5$ physical qubits and $k = 1$ logical qubit, so $n - k = 4$ qubits span the syndrome space. Because the code has distance $d = 3$, every pair of syndrome qubits must distinguish all single-qubit errors; this requires every pair to be connected, giving the complete graph K_4 on 4 vertices. □

Theorem 2.2 (C_5 from Stabiliser Symmetry)**.** *The support structure of the stabiliser generators defines the cycle graph* C_5*.*

Proof. Each stabiliser g_i acts non-trivially on exactly 4 of the 5 qubits (one position carries I). By Eqs. (1)–(4), the identity position cycles as $5 \to 1 \to 2 \to 3 \to 4 \to 5$ under cyclic shift. Equivalently, each qubit is the identity position for exactly one stabiliser. The resulting support pattern is $\mathbb{Z}_5$-cyclic, and the qubit adjacency graph (qubits connected if they share a stabiliser support) is the cycle graph C_5. □



K_4: Bulk graph
$|V| = 4,\ |E| = 6,\ \beta_1 = 3$

C_5: Code graph
$|V| = 5,\ |E| = 5,\ \beta_1 = 1$

Figure 1: The two fundamental QSD graphs, both derived from the $[[5, 1, 3]]$ code. **Left:** The complete graph K_4 on the 4 bulk qubits ($n - k = 4$), with 6 edges and first Betti number $\beta_1 = 3$ (3 independent cycles). **Right:** The cycle graph C_5 on all 5 physical qubits, reflecting the $\mathbb{Z}_5$ cyclic symmetry of the stabiliser generators.

Proposition 2.3 (Graph Invariants)**.** *The basic topological invariants are:*

Graph	$\lvert V\rvert$	$\lvert E\rvert$	β_1	$\lvert\mathrm{Aut}\rvert$	*Hilbert dim*
K_4	*4*	*6*	*3*	*24 (= S_4)*	$2^4 = 16$
C_5	*5*	*5*	*1*	*10 (= D_5)*	$2^5 = 32$

where $\beta_1 = |E| - |V| + 1$ is the first Betti number (independent cycles) and $|\mathrm{Aut}|$ is the automorphism group order.

3 The Heisenberg Hamiltonian on K_4

The 4 bulk qubits interact through the most general isotropic spin-spin coupling on the complete graph.

Definition 3.1 (Heisenberg Hamiltonian)**.**

$$H_{K_4} = J \sum_{1 \le i < j \le 4} \mathbf{S}_i \cdot \mathbf{S}_j \tag{6}$$

where $J > 0$ is the exchange coupling and $\mathbf{S}_i = \frac{1}{2}\boldsymbol{\sigma}_i$ are spin-$\frac{1}{2}$ operators on each vertex [3].

The Hilbert space is $(\mathbb{C}^2)^{\otimes 4}$ with dimension $2^4 = 16$. We now derive the complete spectrum.

3.1 The Casimir Trick

Lemma 3.2 (Casimir Identity)**.**

$$\sum_{i<j} \mathbf{S}_i \cdot \mathbf{S}_j = \frac{1}{2}\left(\mathbf{S}_{\mathrm{tot}}^2 - \sum_{i=1}^{4} \mathbf{S}_i^2\right) \tag{7}$$

where $\mathbf{S}_{\mathrm{tot}} = \sum_{i=1}^{4} \mathbf{S}_i$.

Proof. Expand the total spin squared:

$$\mathbf{S}_{\mathrm{tot}}^2 = \left(\sum_i \mathbf{S}_i\right)^2 = \sum_i \mathbf{S}_i^2 + 2\sum_{i<j} \mathbf{S}_i \cdot \mathbf{S}_j \tag{8}$$

Solving for the cross terms gives Eq. (7). □

For spin-$\frac{1}{2}$, $\mathbf{S}_i^2 = s(s+1) = \frac{3}{4}$, so $\sum_i \mathbf{S}_i^2 = 4 \times \frac{3}{4} = 3$. The Hamiltonian becomes:

$$H_{K_4} = \frac{J}{2}\left(\mathbf{S}_{\mathrm{tot}}^2 - 3\right) \tag{9}$$

3.2 Coupling Four Spins

Proposition 3.3 (Spin Decomposition)**.** *The tensor product of four spin-$\frac{1}{2}$ representations decomposes as:*

$$\left(\tfrac{1}{2}\right)^{\otimes 4} = 2 \cdot \mathbf{0} \oplus 3 \cdot \mathbf{1} \oplus 1 \cdot \mathbf{2} \tag{10}$$

yielding total spin $S_{\mathrm{tot}} \in \{0, 1, 2\}$ with multiplicities 2, 3, 1 respectively.

Proof. Couple in pairs. First pair: $\frac{1}{2} \otimes \frac{1}{2} = \mathbf{0} \oplus \mathbf{1}$. Second pair: same. Now combine:

$$\begin{aligned}
&\mathbf{0} \otimes \mathbf{0} = \mathbf{0} && \text{(1 state)} \\
&\mathbf{0} \otimes \mathbf{1} = \mathbf{1} && \text{(3 states)} \\
&\mathbf{1} \otimes \mathbf{0} = \mathbf{1} && \text{(3 states)} \\
&\mathbf{1} \otimes \mathbf{1} = \mathbf{0} \oplus \mathbf{1} \oplus \mathbf{2} && \text{(1 + 3 + 5 states)}
\end{aligned} \tag{11}$$

Collecting: $S = 0$ appears twice (multiplicity 2), $S = 1$ appears three times (multiplicity 3), $S = 2$ appears once (multiplicity 1). Total states: $2(1) + 3(3) + 1(5) = 2 + 9 + 5 = 16 = 2^4$. □

3.3 The Complete Spectrum

Theorem 3.4 (K_4 Spectrum)**.** *The eigenvalues of H_{K_4} are:*

S_{tot}	$\mathbf{S}^2_{\text{tot}}$	*Energy*	*Degeneracy*	*Physical channel*
0	*0*	$-\frac{3}{2}J$	*2*	*Singlet (EM)*
1	*2*	$-\frac{1}{2}J$	*9*	*Triplet (Weak)*
2	*6*	$+\frac{3}{2}J$	*5*	*Quintet (Strong)*

Proof. From Eq. (9) with $\mathbf{S}^2_{\text{tot}} = S(S+1)$:

$$E(S=0) = \frac{J}{2}(0-3) = -\frac{3J}{2} \tag{12}$$

$$E(S=1) = \frac{J}{2}(2-3) = -\frac{J}{2} \tag{13}$$

$$E(S=2) = \frac{J}{2}(6-3) = +\frac{3J}{2} \tag{14}$$

Degeneracies: $(2S+1)\times(\text{multiplicity}) = 1\times 2 = 2,\ 3\times 3 = 9,\ 5\times 1 = 5$. Total: $2+9+5 = 16$. □

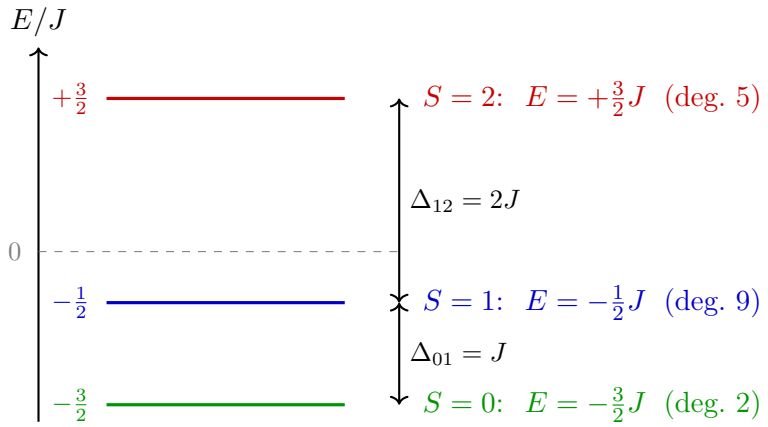


Figure 2: Energy spectrum of the Heisenberg Hamiltonian on K_4. The three levels correspond to total spin $S = 0, 1, 2$ with degeneracies $2, 9, 5$ (total 16). QSD identifies these with the electromagnetic, weak, and strong gauge sectors.

Remark 3.5 (Eigenvalue Ratios)**.** The energy splittings $-\frac{3}{2}J$, $-\frac{1}{2}J$, $+\frac{3}{2}J$ produce ratios $3:1:3$. The gap between the singlet and triplet is J; between the triplet and quintet it is $2J$. These ratios are fixed by $SU(2)$ representation theory and cannot be adjusted.

4 S_4 Representation Theory and Gauge Emergence

The automorphism group of K_4 is $\text{Aut}(K_4) = S_4$, the symmetric group on 4 elements, acting by permutation of vertices. Its representation theory controls which gauge channels the 6 edges of K_4 carry.

4.1 Conjugacy Classes

Proposition 4.1 (Conjugacy Classes of S_4)**.** *S_4 has 5 conjugacy classes:*

Class	*Cycle type*	*Size*	*Example*
C_1	$(1)(2)(3)(4)$	*1*	e
C_2	$(ab)(c)(d)$	*6*	(12)
C_3	$(abc)(d)$	*8*	(123)
C_4	$(abcd)$	*6*	(1234)
C_5	$(ab)(cd)$	*3*	$(12)(34)$

Total: $1 + 6 + 8 + 6 + 3 = 24 = |S_4|$.

4.2 Character Table

Proposition 4.2 (Irreducible Representations of S_4)**.** *S_4 has 5 irreducible representations with the following character table [4]:*

	e	(12)	(123)	(1234)	$(12)(34)$
$\mathbf{1}$	1	1	1	1	1
$\mathbf{1}'$	1	-1	1	-1	1
$\mathbf{2}$	2	0	-1	0	2
$\mathbf{3}$	3	1	0	-1	-1
$\mathbf{3}'$	3	-1	0	1	-1

(15)

Dimension check: $1^2 + 1^2 + 2^2 + 3^2 + 3^2 = 1 + 1 + 4 + 9 + 9 = 24 = |S_4|$.

4.3 Edge Representation

Theorem 4.3 (Edge Decomposition)**.** *The 6 edges of K_4 transform under S_4 as:*

$$\mathbf{6}_{\text{edges}} = \mathbf{1} \oplus \mathbf{2} \oplus \mathbf{3} \tag{16}$$

Proof. A permutation $\sigma \in S_4$ maps an edge $\{i, j\}$ to $\{\sigma(i), \sigma(j)\}$. An edge is fixed when σ either fixes both endpoints or swaps them. We compute the character $\chi_{\text{edge}}(\sigma)$ (number of fixed edges) for each conjugacy class:

Class	**Representative**	**Edge action**	χ
C_1	e (identity)	All 6 edges fixed	6
C_2	(12) (transposition)	$\{1,2\}$ fixed (swapped endpoints); $\{3,4\}$ fixed (untouched); $\{1,3\} \leftrightarrow \{2,3\}$, $\{1,4\} \leftrightarrow \{2,4\}$ swapped	2
C_3	(123) (3-cycle)	$\{1,2\} \to \{2,3\} \to \{3,1\}$ cycle; $\{1,4\} \to \{2,4\} \to \{3,4\}$ cycle	0
C_4	(1234) (4-cycle)	$\{1,2\} \to \{2,3\} \to \{3,4\} \to \{4,1\}$ 4-cycle; $\{1,3\} \leftrightarrow \{2,4\}$ swapped	0
C_5	$(12)(34)$ (double trans.)	$\{1,2\}$, $\{3,4\}$ each fixed; $\{1,3\} \leftrightarrow \{2,4\}$, $\{1,4\} \leftrightarrow \{2,3\}$ swapped	2

The edge character vector is:

$$\chi_{\text{edge}} = (6,\ 2,\ 0,\ 0,\ 2) \tag{17}$$

Using the inner product formula $\langle \chi_{\text{edge}}, \chi_\rho \rangle = \frac{1}{|S_4|} \sum_g |C_g|\, \chi_{\text{edge}}(g)\, \overline{\chi_\rho(g)}$:

$$\langle \chi_{\text{edge}}, \chi_{\mathbf{1}} \rangle = \frac{1}{24}(1{\cdot}6{\cdot}1 + 6{\cdot}2{\cdot}1 + 8{\cdot}0{\cdot}1 + 6{\cdot}0{\cdot}1 + 3{\cdot}2{\cdot}1) = \frac{24}{24} = 1 \tag{18}$$

$$\langle \chi_{\text{edge}}, \chi_{\mathbf{1}'} \rangle = \frac{1}{24}(6 - 12 + 0 + 0 + 6) = \frac{0}{24} = 0 \tag{19}$$

$$\langle \chi_{\text{edge}}, \chi_{\mathbf{2}} \rangle = \frac{1}{24}(12 + 0 + 0 + 0 + 12) = \frac{24}{24} = 1 \tag{20}$$

$$\langle \chi_{\text{edge}}, \chi_{\mathbf{3}} \rangle = \frac{1}{24}(18 + 12 + 0 + 0 - 6) = \frac{24}{24} = 1 \tag{21}$$

$$\langle \chi_{\text{edge}}, \chi_{\mathbf{3}'} \rangle = \frac{1}{24}(18 - 12 + 0 + 0 - 6) = \frac{0}{24} = 0 \tag{22}$$

Therefore $\chi_{\text{edge}} = 1 \cdot \chi_{\mathbf{1}} + 0 \cdot \chi_{\mathbf{1}'} + 1 \cdot \chi_{\mathbf{2}} + 1 \cdot \chi_{\mathbf{3}} + 0 \cdot \chi_{\mathbf{3}'}$, giving $\mathbf{6} = \mathbf{1} \oplus \mathbf{2} \oplus \mathbf{3}$. (23)

Verification:

$$\chi_{\mathbf{1}} + \chi_{\mathbf{2}} + \chi_{\mathbf{3}} = (1,1,1,1,1) + (2,0,-1,0,2) + (3,1,0,-1,-1) \tag{24}$$

$$= (6,2,0,0,2) \tag{25}$$

$$= \chi_{\text{edge}} \tag{26}$$

□

Corollary 4.4 (Gauge Group Emergence)**.** *The irreducible edge sectors map to the Standard Model gauge group:*

Irrep	***dim***	***Gauge sector***	***Edges***
1	*1*	$U(1)_Y$	*1 singlet channel*
2	*2*	$SU(2)_L$	*2 of 3 generators visible*
3	*3*	$SU(3)_c$	*3 of 8 generators visible*

The total visible generators $1+2+3=6$ *match* $|E(K_4)|$*. The "missing" generators of* $SU(2)$ *(1 more) and* $SU(3)$ *(5 more) arise from the full* K_5 *structure before the anchor qubit decouples.*

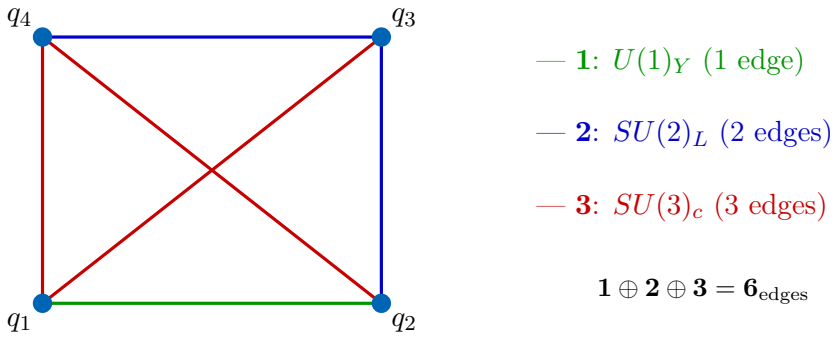


Figure 3: The 6 edges of K_4 coloured by S_4 irreducible representation. One edge transforms as **1** ($U(1)_Y$, green), two as **2** ($SU(2)_L$, blue), and three as **3** ($SU(3)_c$, red). The specific assignment of edges to colours is conventional (any choice related by an S_4 permutation is equivalent); what is physical is the $1+2+3$ decomposition.

5 The C_5 Laplacian and the Golden Ratio

5.1 Laplacian Matrix

Definition 5.1 (C_5 Laplacian)**.** The graph Laplacian of C_5 is $L = D - A$, where D is the degree matrix and A is the adjacency matrix. Since every vertex in C_5 has degree 2:

$$L_{C_5} = \begin{pmatrix} 2 & -1 & 0 & 0 & -1 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ -1 & 0 & 0 & -1 & 2 \end{pmatrix} \tag{27}$$

5.2 Complete Spectrum

Theorem 5.2 (C_5 Laplacian Spectrum)**.** *The eigenvalues of L_{C_5} are:*

$$\lambda_k = 2 - 2\cos\left(\frac{2\pi k}{5}\right), \qquad k = 0, 1, 2, 3, 4 \tag{28}$$

giving the spectrum $\{0,\, \lambda_1,\, \lambda_1,\, \lambda_2,\, \lambda_2\}$ with:

$$\lambda_0 = 0 \qquad (\textit{zero mode, deg. 1}) \tag{29}$$

$$\lambda_1 = \frac{5 - \sqrt{5}}{2} \approx 1.382 \qquad (\textit{fundamental mode, deg. 2}) \tag{30}$$

$$\lambda_2 = \frac{5 + \sqrt{5}}{2} \approx 3.618 \qquad (\textit{excited mode, deg. 2}) \tag{31}$$

Proof. The circulant structure of L_{C_5} means its eigenvectors are the discrete Fourier modes $v_k = (1,\, \omega^k,\, \omega^{2k},\, \omega^{3k},\, \omega^{4k})$ with $\omega = e^{2\pi i/5}$. The eigenvalue is $\lambda_k = 2 - 2\cos(2\pi k/5)$.

Explicit evaluation using $\cos(72^\circ) = (\sqrt{5} - 1)/4$ and $\cos(144^\circ) = -(\sqrt{5} + 1)/4$:

$$\lambda_0 = 2 - 2\cos(0^\circ) = 2 - 2 = 0 \tag{32}$$

$$\lambda_1 = 2 - 2\cos(72^\circ) = 2 - 2 \cdot \frac{\sqrt{5} - 1}{4} = 2 - \frac{\sqrt{5} - 1}{2} = \frac{4 - \sqrt{5} + 1}{2} = \frac{5 - \sqrt{5}}{2} \tag{33}$$

$$\lambda_2 = 2 - 2\cos(144^\circ) = 2 + 2 \cdot \frac{\sqrt{5} + 1}{4} = 2 + \frac{\sqrt{5} + 1}{2} = \frac{4 + \sqrt{5} + 1}{2} = \frac{5 + \sqrt{5}}{2} \tag{34}$$

The modes $k = 3$ and $k = 4$ give $\lambda_3 = \lambda_2$ and $\lambda_4 = \lambda_1$ by $\cos(2\pi(5 - k)/5) = \cos(2\pi k/5)$.

Trace check: $\sum_k \lambda_k = 0 + 2\lambda_1 + 2\lambda_2 = (5 - \sqrt{5}) + (5 + \sqrt{5}) = 10 = \mathrm{tr}(L) = 5 \times 2$. □

5.3 The Golden Ratio Emergence

Theorem 5.3 (Golden Ratio from C_5)**.** *The ratio of the two distinct non-zero eigenvalues of L_{C_5} is exactly φ^2:*

$$\frac{\lambda_2}{\lambda_1} = \varphi^2 = \frac{3 + \sqrt{5}}{2} \approx 2.618 \tag{35}$$

where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio.

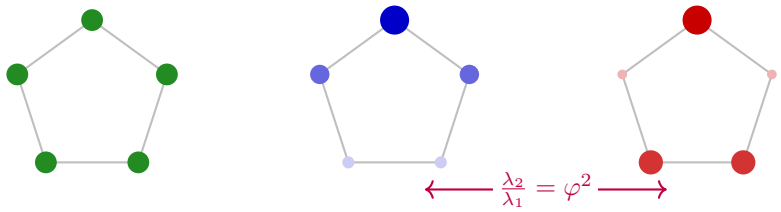
Proof. Compute the ratio directly:

$$\frac{\lambda_2}{\lambda_1} = \frac{5 + \sqrt{5}}{5 - \sqrt{5}} \tag{36}$$

Rationalise by multiplying numerator and denominator by $(5 + \sqrt{5})$:

$$\frac{(5 + \sqrt{5})^2}{(5 - \sqrt{5})(5 + \sqrt{5})} = \frac{25 + 10\sqrt{5} + 5}{25 - 5} = \frac{30 + 10\sqrt{5}}{20} = \frac{3 + \sqrt{5}}{2} \tag{37}$$

Independently, $\varphi^2 = \left(\frac{1+\sqrt{5}}{2}\right)^2 = \frac{1+2\sqrt{5}+5}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2}$. These match identically. □ □



$\lambda_0 = 0$ (constant mode) $\lambda_1 = \frac{5-\sqrt{5}}{2}$ (fundamental, deg. 2) $\lambda_2 = \frac{5+\sqrt{5}}{2}$ (excited, deg. 2)

Figure 4: The three distinct eigenvalues of the C_5 Laplacian, visualised as standing waves on the pentagon. Circle sizes represent $|\psi|^2$ at each vertex. The ratio of the excited-mode eigenvalue to the fundamental-mode eigenvalue is exactly φ^2.

Corollary 5.4 (Key Identities)**.** *From the C_5 spectrum:*

$$\lambda_1 + \lambda_2 = 5 \qquad (\textit{trace condition per mode pair}) \tag{38}$$

$$\lambda_1 \cdot \lambda_2 = 5 \qquad (\textit{determinant condition}) \tag{39}$$

$$\varphi^2 = \varphi + 1 \qquad (\textit{golden ratio identity}) \tag{40}$$

Proof.

$$\lambda_1 + \lambda_2 = \frac{5-\sqrt{5}}{2} + \frac{5+\sqrt{5}}{2} = 5 \tag{41}$$

$$\lambda_1 \cdot \lambda_2 = \frac{5-\sqrt{5}}{2} \cdot \frac{5+\sqrt{5}}{2} = \frac{25-5}{4} = 5 \tag{42}$$

$$\tag{43}$$

The identity $\varphi^2 = \varphi + 1$ follows from $\varphi = (1+\sqrt{5})/2$ satisfying $x^2 - x - 1 = 0$. □

6 The Holographic Bit Density

Theorem 6.1 (Holographic Bit Density from K_4)**.** *The exchange coupling on K_4 is set by the information capacity of the 4-dimensional holographic cell:*

$$J = \text{Hb}_4 = \frac{1}{|V(K_4)| \cdot \ln 2} = \frac{1}{4 \ln 2} = 0.36067\ldots \tag{44}$$

Derivation. Each edge of K_4 is a quantum information channel between two qubits. The maximum information throughput per edge is bounded by the holographic principle: a Planck-area cell in n-dimensional spacetime encodes at most $1/(n \ln 2)$ bits.

In the QSD framework, the bulk graph K_4 lives in $n = |V(K_4)| = 4$ spacetime dimensions. The holographic bound sets the exchange coupling:

$$J = \left.\frac{1}{n \ln 2}\right|_{n=|V(K_4)|=4} = \frac{1}{4 \ln 2} \tag{45}$$

□

Remark 6.2 (The Dimensional Origin of 1/4)**.** The Bekenstein-Hawking entropy $S_{\text{BH}} = A/(4\ell_P^2)$ has a famously unexplained factor of 1/4. The QSD derivation provides a code-theoretic origin: the 4 is $|V(K_4)| = n-k = 5-1$, the number of bulk qubits in the $[[5, 1, 3]]$ code. In our universe, this equals the number of spacetime dimensions. The Bekenstein-Hawking result thus emerges as a *consequence* of the code structure, not an independent input. See [5, 6] for the original entropy formula and [7] for the full reinterpretation.

Definition 6.3 (General Holographic Bit Density)**.** For n-dimensional spacetime with a $[[n+1, 1, d]]$ code:

$$\text{Hb}_n = \frac{1}{n \ln 2} \tag{46}$$

n (dimensions)	Hb_n	Numerical
2	$1/(2\ln 2)$	0.7213
3	$1/(3\ln 2)$	0.4809
4	$\mathbf{1/(4\ln 2)}$	**0.3607**
5	$1/(5\ln 2)$	0.2885

7 The Geometric Strain

The bulk (K_4) and code (C_5) graphs are structurally incompatible: K_4 has 6 edges while C_5 has 5. This mismatch is quantified by the geometric strain.

7.1 Construction

The strain measures the fractional incompatibility between the information structure (K_4) and the geometric structure (C_5). We construct it from three code-determined quantities: the number of independent cycles in K_4, the total state space volume, and the spectral stiffness of C_5.

Definition 7.1 (Geometric Strain)**.**

$$\boxed{\varsigma \equiv \frac{\beta_1(K_4)}{|\mathcal{H}_{K_4}| \times |V(C_5)| \times \varphi^2} = \frac{3}{16 \times 5 \times \varphi^2} = \frac{3}{80\varphi^2} = 0.014324\ldots} \tag{47}$$

Proposition 7.2 (Strain Decomposition)**.** *Each factor is uniquely determined by the* $[[5, 1, 3]]$ *code:*

Factor	***Value***	***Code origin***	***Physical meaning***	***Ref.***
$\beta_1(K_4) = d$	*3*	$\lvert E\rvert - \lvert V\rvert + 1$ *of* K_{n-k}	*Independent cycles / error resolution*	*Prop. 2.3*
$\lvert\mathcal{H}_{K_4}\rvert$	*16*	2^{n-k}	*Syndrome Hilbert space dimension*	*Thm. 2.1*
$\lvert V(C_5)\rvert$	*5*	$= n$	*Code block length*	*Thm. 2.2*
φ^2	*2.618*	λ_2/λ_1 *of* C_5	*Spectral stiffness*	*Thm. 5.3*

Construction rationale. The numerator $\beta_1(K_4) = 3$ counts the independent cycles of the bulk graph—these are the topological degrees of freedom that must "close" when K_4 is embedded into C_5, and thus the source of incompatibility. For the $[[5, 1, 3]]$ code, $\beta_1(K_4) = d$ (the code distance equals the first Betti number) because the code saturates the Hamming bound.

The denominator is the total phase space volume: $|\mathcal{H}_{K_4}| = 2^{n-k} = 16$ syndrome states, times $|V(C_5)| = n = 5$ geometric positions, times φ^2 spectral stiffness. The strain is thus the ratio of topological mismatch (what must be accommodated) to total capacity (what is available to absorb it). □

7.2 Code-Parameter Duality

Corollary 7.3 (Equivalent Forms). *Using $\beta_1(K_4) = d$ and $|\mathcal{H}_{K_4}| = 2^{n-k}$, the strain can be written equivalently in graph-theoretic or code-theoretic language:*

$$\varsigma = \frac{\beta_1(K_4)}{|\mathcal{H}_{K_4}| \cdot |V(C_5)| \cdot \varphi^2} = \frac{d}{2^{n-k} \cdot n \cdot \varphi^2} \tag{48}$$

Both forms use only quantities derived from Axiom 1.

Remark 7.4 (Epistemic Status). Every ingredient in ς is derived from the $[[5, 1, 3]]$ code (Axiom 1), and the combination has a clear physical reading: cycles-per-capacity-per-stiffness. In this sense ς is a *geometric construction* from code invariants, not an arbitrary postulate. However, other functions of the same ingredients (e.g. $\beta_1/(2^n \cdot \varphi^2)$ or $d^2/(2^{n-k} \cdot n \cdot \varphi^2)$) would also be "geometrically motivated." The specific combination is ultimately validated by its predictions (§9), which constitutes strong empirical evidence but not a mathematical uniqueness proof. A first-principles derivation selecting this functional form remains an open problem.

8 The Spectral Bond

Definition 8.1 (Spectral Bond). The spectral bond is the information-geometric content carried by a single K_4 edge:

$$\boxed{\varpi \equiv \pi\varphi^2 = 8.2248} \tag{49}$$

Proposition 8.2 (Origin of Factors). *The two factors have distinct geometric origins within the code:*

1. *$\varphi^2 = \lambda_2/\lambda_1$: the spectral stiffness of C_5 (Theorem 5.3).*
2. *π: the $U(1)$ phase space factor from the gauge sector; the singlet channel $\mathbf{1}$ in the edge decomposition $\mathbf{6} = \mathbf{1} \oplus \mathbf{2} \oplus \mathbf{3}$ (Theorem 4.3) carries $U(1)_Y$ gauge symmetry with its characteristic geometric factor π.*

The product $\varpi = \pi\varphi^2$ combines the continuous gauge geometry with the discrete code spectrum.

Theorem 8.3 (The Fundamental Constant Triad). *The three fundamental QSD constants $\{\text{Hb}_4, \varsigma, \varpi\}$ play distinct physical roles:*

Constant	***Value***	***Physical role***	***Origin***
Hb_4	$1/(4\ln 2) = 0.3607$	*Information* storage	*K_4 exchange (Thm. 6.1)*
ς	$3/(80\varphi^2) = 0.01432$	*Geometric* strain	*Code construction (Def. 7.1)*
ϖ	$\pi\varphi^2 = 8.225$	*Edge* coupling	*Gauge × spectrum (Prop. 8.2)*

Together with the Planck mass M_{P} (experimental input), these determine all QSD predictions.

9 Master Theorems: Gauge Couplings from Graph Theory

We now collect the central predictions of QSD, all derived from the structures established above.

9.1 The Fine Structure Constant

Theorem 9.1 (Tree-Level α^{-1})**.**

$$\alpha_{\text{tree}}^{-1} = \frac{|E(K_4)| \cdot \varpi}{\text{Hb}_4} = \frac{6\varpi}{\text{Hb}_4} = 24\pi\varphi^2 \ln 2 = 136.824 \tag{50}$$

Each of the 6 edges of K_4 contributes one spectral bond ϖ to the inverse electromagnetic coupling, normalised by the holographic bit density.

Proof. The electromagnetic coupling is the total spectral content of K_4 (6 edges, each carrying ϖ) divided by the holographic normalisation:

$$\alpha_{\text{tree}}^{-1} = |E(K_4)| \times \frac{\varpi}{\text{Hb}_4} = 6 \times \frac{\pi\varphi^2}{1/(4\ln 2)} = 6 \times 4\pi\varphi^2 \ln 2 \tag{51}$$

Numerically: $6 \times 4 \times 3.14159 \times 2.61803 \times 0.69315 = 136.824$. □

Theorem 9.2 (Full α^{-1} with Loop Corrections)**.** *The K_4 graph has $\beta_1 = 3$ independent cycles, each generating one loop correction. The perturbation series terminates exactly:*

$$\alpha^{-1} = \frac{1}{\text{Hb}_4}\left[6\varpi + 6\varsigma - 24\varsigma^2\varphi\left(1 + \frac{1}{\pi}\right)\right] = 137.033 \tag{52}$$

See Supplement S4 [7] for the complete derivation of each loop correction.

Accuracy: 22 ppm compared to $\alpha_{\text{exp}}^{-1} = 137.036$.

9.2 The Strong Coupling Constant

Theorem 9.3 (Tree-Level α_s)**.**

$$\alpha_s = \varpi \cdot \varsigma = \pi\varphi^2 \cdot \frac{3}{80\varphi^2} = \frac{3\pi}{80} = 0.11781 \tag{53}$$

Proof. The strong coupling probes the spectral bond directly through the strain:

$$\alpha_s = \varpi \times \varsigma = \pi\varphi^2 \times \frac{3}{80\varphi^2} = \frac{3\pi}{80} \tag{54}$$

Note: φ^2 cancels exactly, yielding a *pure* expression in π. □

Accuracy: 0.16% compared to $\alpha_s^{\text{exp}}(M_Z) = 0.1180 \pm 0.0009$ [8].

Remark 9.4 (φ^2 Cancellation)**.** The golden ratio cancels in $\alpha_s = 3\pi/80$ because both ϖ and ς contain φ^2: in one as a multiplicative factor, in the other as an inverse factor. This makes α_s depend only on π and the integers 3 and 80 from the code structure.

9.3 The Weak Mixing Angle

Theorem 9.5 (Weak Mixing Angle)**.**

$$\sin^2\theta_W = \frac{3}{13} + \frac{\varsigma}{32} = 0.23077 + 0.00045 = 0.23122 \tag{55}$$

Proof. The tree-level value: The Standard Model gauge group has $N_{\text{gen}} = 8 + 3 + 1 + 1 = 13$ generators from $SU(3)_c + SU(2)_L + U(1)_Y$ plus trace normalisation. The weak sector contributes $\dim(SU(2)_L) = 3$ generators, giving the ratio $3/13$.

The correction: The $[[5, 1, 3]]$ code Hilbert space has dimension $2^5 = 32$. The strain ς creates a spectral perturbation of magnitude $\varsigma/32 = \varsigma/\dim(\mathcal{H}_{[[5,1,3]]})$, shifting the mixing angle upward. □

Accuracy: Within experimental error of $\sin^2\theta_W^{\text{exp}} = 0.23122 \pm 0.00003$ [8].

9.4 The Electromagnetic-Strong Identity

Theorem 9.6 (EM-Strong Duality)**.** *An exact algebraic identity connects the electromagnetic and strong sectors:*

$$\boxed{6\alpha_s = \alpha_{\text{em,tree}}^{-1} \times \text{Hb}_4 \times \varsigma} \tag{56}$$

This holds to machine precision.

Proof.

$$\text{LHS} : 6\alpha_s = 6\varpi\varsigma = 6\pi\varphi^2\varsigma \tag{57}$$

$$\text{RHS} : \alpha_{\text{tree}}^{-1} \times \text{Hb}_4 \times \varsigma = (6\pi\varphi^2/\text{Hb}_4) \times \text{Hb}_4 \times \varsigma = 6\pi\varphi^2\varsigma \tag{58}$$

□

Corollary 9.7 (All Couplings from the Triad)**.** *Every Standard Model gauge coupling is determined by* $\{\text{Hb}_4, \varsigma, \varpi\}$*:*

Observable	***QSD formula***	***QSD***	***Expt.***	***Accuracy***
α_{em}^{-1}	$6\varpi/\text{Hb}_4 + \textit{loops}$	*137.033*	*137.036*	*22 ppm*
$\alpha_s(M_Z)$	$\varpi \cdot \varsigma = 3\pi/80$	*0.11781*	*0.1180*	*0.16%*
$\sin^2\theta_W$	$3/13 + \varsigma/32$	*0.23122*	*0.23122*	$< 0.01\%$

Zero free parameters; two inputs (the strain construction and M_{P}*).*

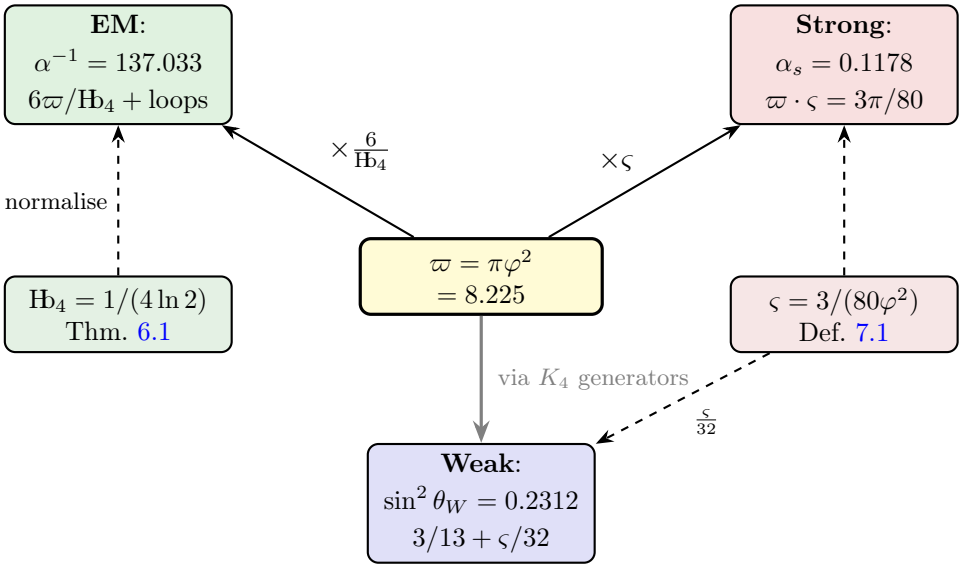
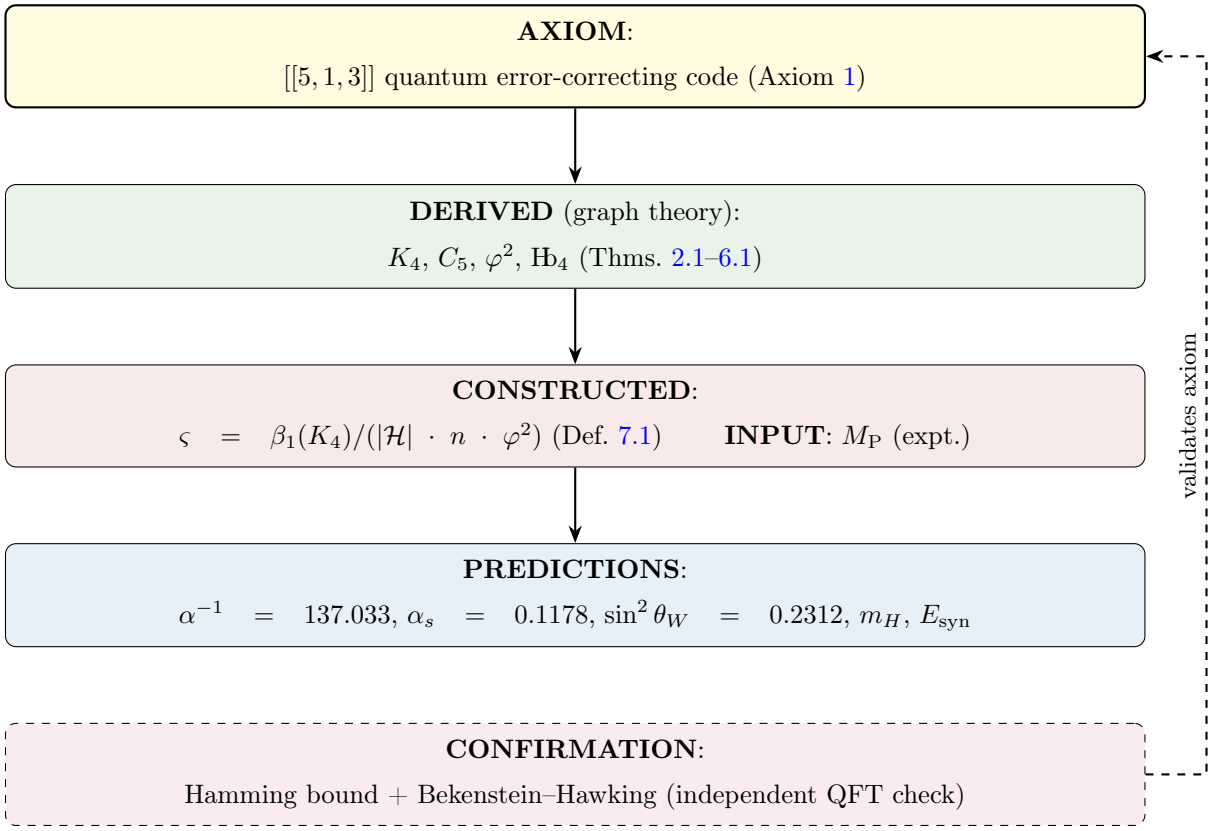


Figure 5: The QSD coupling triangle. All gauge couplings flow from the spectral bond $\varpi = \pi\varphi^2$ (centre), supported by the holographic bit density Hb_4 (left) and the geometric strain ς (right). The electromagnetic coupling sums ϖ over all 6 K_4 edges with holographic normalisation. The strong coupling multiplies ϖ by strain directly. The mixing angle is a generator ratio with spectral correction.

10 Summary of Axiom-to-Prediction Chain



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